HW1: 3D Scene Reconstruction Report

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1. Implementation

1.1 Reconstruction pipeline

Each RGB-D frame is converted into a point cloud. I register it to the previous frame, chain the results to obtain the camera path, and merge all clouds into a single map.

1. **Depth unprojection (written from scratch).** I read width, height and hfov from the capture's intrinsics.json. With a pinhole model and square pixels, fx = fy = (W/2) / tan(hfov/2), which is 256 for 512 px and 90 degrees, and cx = W/2, cy = H/2. Depth PNGs are uint16 in mm, so I divide by 1000. For each pixel with z > 0: x = (u - cx) z / fx and y = (v - cy) z / fy. Pixels with zero depth are dropped. Colours are changed from BGR to RGB. The camera axes are +X right, +Y down, +Z forward.

2. **Preprocessing.** Voxel downsampling with voxel size v, then normals (radius 2v, 30 neighbours) and FPFH features (radius 5v, 100 neighbours).

3. **Global registration.** RANSAC on FPFH matches with a mutual filter, 3 points per sample, distance gate 1.5v, edge length (0.9) and distance checkers, at most 100000 iterations with 0.999 confidence. The seed is fixed for every pair so that runs repeat on the same machine.

4. **Local refinement.** Open3D point-to-plane ICP starting from the RANSAC pose, threshold 1.5v, 100 iterations.

5. **Second ICP start.** I also run ICP from the identity. I keep the result with the smaller translation, unless the larger one is clearly better: fitness higher by more than 0.02 and inlier RMSE at most 1.1 times worse.

6. **Yaw-only constraint.** The Habitat agent only turns left and right, so pitch and roll between frames are always zero. I refine the chosen pose by repeating one ICP step and then projecting the rotation back to a rotation about camera Y, up to 30 times.

7. **Pose chaining.** The result maps current to previous, so T_i = T_(i-1) * T_pair. The camera center is the translation part of T_i.

8. **Map and evaluation.** Each cloud is downsampled to 2 cm, transformed by T_i and merged, and the provided function removes the ceiling. Mean L2 is the mean distance between the estimated camera centres and the GT positions relative to the first GT pose. The estimated path is drawn in red and GT in black.

Final settings: voxel 8 cm, ICP threshold and RANSAC gate 12 cm, normal radius 16 cm, FPFH radius 40 cm, map voxel 2 cm, seed 0.

1.2 Problems I met and how I fixed them

Problem	What I saw	Fix	Why
180 degree flip, 3.17 m (floor 1, pair 103 to 104)	A 10 degree turn in front of a blank wall with noisy depth. RANSAC gave a flipped pose and ICP kept it.	Also run ICP from the identity	Neighbouring frames are close, so identity is a good second guess and a flip fits worse.
Slide along a wall, 0.274 m (floor 1, pair 104 to 105)	The slid pose had higher fitness but also higher RMSE. Choosing by fitness alone failed for 14 of 30 seeds.	Keep the smaller motion unless the larger one is better in fitness and RMSE	RMSE helps separate a real alignment from a slide. Fitness alone does not.
Roll error, 0.300 m (floor 2, frames 165-171)	The camera was 0.5 m from a flat white wall. Each pair got up to about 7 degrees of roll error, about 13 degrees in total, and the rest of the map was tilted.	Yaw-only ICP	A flat wall cannot tell how much the camera rotated around the wall's normal, but the agent cannot roll anyway.

Different result on Mac and server	Open3D 0.18 and 0.20 sample RANSAC differently. The same settings gave 0.274 m and 0.091 m.	Use the rules above and test several seeds	A result should not depend on one random draw.

Voxel size. I tested sizes on the lab server. On floor 1 with the first identity check: 3 cm 0.345, 4 cm 0.212, 5 cm 0.243, 6 cm 0.212, 8 cm 0.091, 10 cm 0.372 m. On floor 2 with the final pipeline: 5 cm 0.208, 6 cm 0.168, 8 cm 0.154, 10 cm 3.49 m. I chose 8 cm based on GT error on these same captures, so these are tuned values. With the default 5 cm, the final pipeline gives 0.240 m on floor 1 and 0.208 m on floor 2.

2. Data verification

2.1 Measurements

All eight factors are recorded in the first-pass run at that run's voxel size. RawPointCount and VoxelizedPointYield are per image. CoarseRANSACFitness, CoarseRANSACRMSE, CoarseRANSACCorrespondences, ICPInlierFitness, ICPInlierRMSE and ICPInlierCorrespondences are per adjacent pair.

Sources: docs/factors.md and docs/e2e-workflow.md in the course repo, and Open3D RegistrationResult (https://www.open3d.org/docs/release/python_api/open3d.pipelines.registration.RegistrationResult.html). Plots are in experiments/<name>/plots/.

2.2 Phase 1: floor 1, provided capture, 191 frames

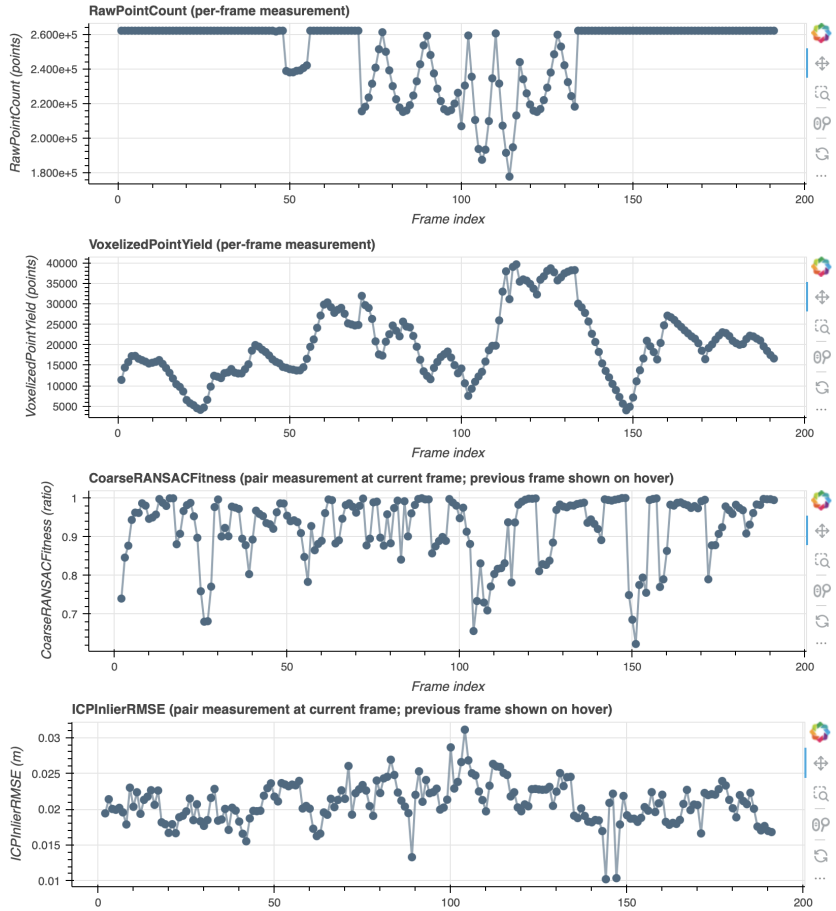


Figure 1. f1_baseline first-pass timelines at 5 cm voxel: RawPointCount, VoxelizedPointYield, CoarseRANSACFitness, ICPInlierRMSE.

- RawPointCount drops at frames 49-55 and 71-133 because depth returns are missing.
- VoxelizedPointYield is lowest at close views, around frames 20-25 and 147-149. It is about 4000 at 5 cm (Figure 1) and about 1500-2500 at 8 cm.

- Fitness is lowest at frames 26-27, 108, and 150-151, but I checked these pairs against the GT, and they are correct (within 2 cm and 0.5 degrees).
- The real failures are at frames 103-106, a blank noisy wall during a turn. In the f1_yaw run, frames 105 and 104 have the highest ICPInlierRMSE of all 190 pairs.

So low fitness means small overlap, not a wrong registration. Missing depth alone is also not the cause: frame 114 has the most missing depth (32%) and it still registers correctly.

2.3 Phase 2: floor 2, my own capture

Capture path: eval/_data/second_floor/314540056. 314 frames, 32.1 m path, steps of 0.2 m and 6 degrees. The largest turn between two frames is 18 degrees.

- Route (frames are approximate): started in the first bedroom and looked around (1-90), went to the first bathroom (around 106), back to the hallway (around 121), then to the second bathroom where I made the wall turn (136-200), along the hallway to the top of the stairs, where I looked into the north bedroom from its door and looked at the stairs without walking on them (217-275), and ended in the last bedroom (286-314).
- Zones: I entered benign (frames 142-200) and moderate (217-275). I did not enter severe, so no severe frames are in this capture. I saw the north room near that zone only from its doorway.
- My first capture had 295 frames, but I turned up to 162 degrees between two saved frames. 8 of 294 pairs failed (7 after turns of 24-162 degrees, one after a 6-degree turn with a 0.35 m move), and Mean L2 was 8.48 m on the server with the motion-prior pipeline. I recorded the capture again with smaller turns between frames.
- Coverage check: I compared the GT-posed depth with the Replica mesh, only as a check and not in the pipeline. 91% of the floor surface is within 15 cm of the captured depth, and 95% of the floor plan cells were seen. A few room corners were missed.

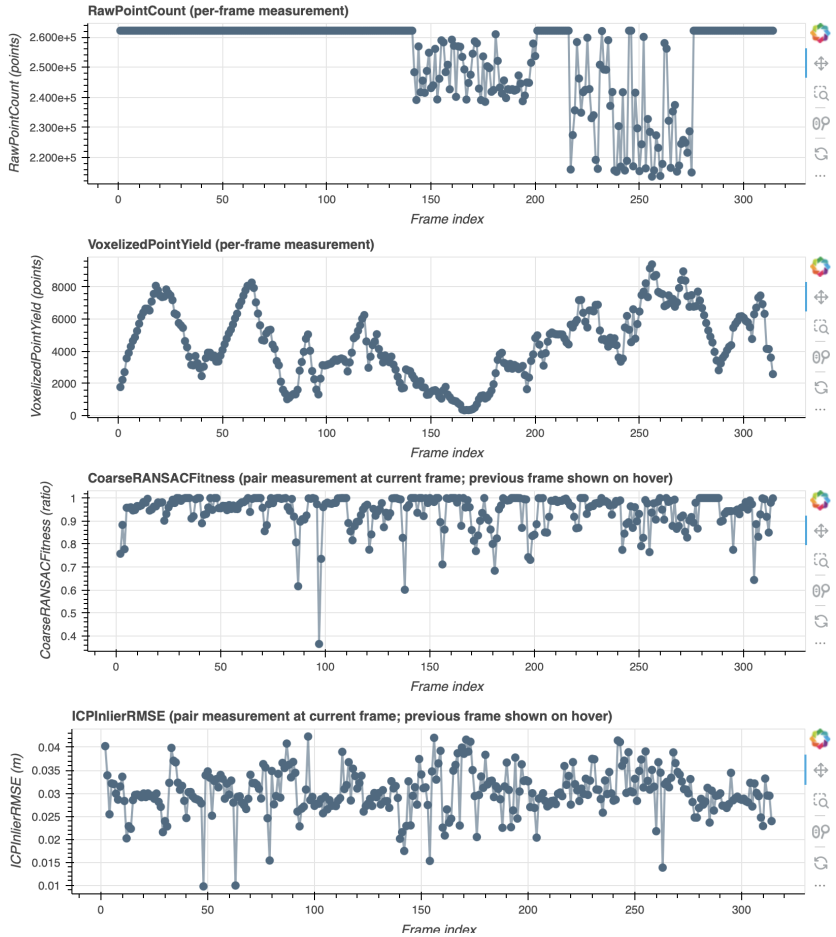


Figure 2. f2_yaw first-pass timelines at 8 cm voxel: RawPointCount, VoxelizedPointYield, CoarseRANSACFitness, ICPInlierRMSE.

- RawPointCount falls inside the zones, down to about 215000 of 262144 in the moderate zone, but those pairs still register well.
- VoxelizedPointYield drops to about 320-500 at frames 165-171. This is the turn in front of the flat white wall, in the benign zone. Most of the trajectory error starts here.
- Only yield separates these frames. Their ICP fitness is high (mean 0.92) and their RMSE is mixed: frame 171 ranks 3 of 313 but frame 168 ranks 290. My prediction that RMSE would peak at this wall was only partly right.
- The lowest fitness on this floor is at frames 97 (0.37) and 138 (0.60), and both pairs are correct (about 1 cm and 0.5 degrees).

2.4 What I changed based on the evidence

Frame selection with SPARQL. On floor 1, I dropped frames where the pair had CoarseRANSACFitness below 0.70. This removed frames 26, 27, 108 and 151:

```
SELECT DISTINCT ?frameIndex WHERE {
  exp:f1_yaw hw1:onBatch ?batch .
  ?batch hw1:hasFrame ?frame .
  ?frame hw1:frameIndex ?frameIndex ; hw1:hasDepthImage ?depth .
  FILTER NOT EXISTS {
    ?pOcc a hw1:DepthPairFactor ; hw1:inExperiment exp:f1_yaw ;
          hw1:hasCurrentFrame ?depth ; hw1:value ?pv ;
          hw1:hasDefinition hw1:CoarseRANSACFitness .
    FILTER (?pv < 0.70)
  }
} ORDER BY ?frameIndex
```

On floor 2, I used the same query shape with hw1:SingleImageFactor and VoxelizedPointYield < 500. This removed the wall frames 165-171.

Pipeline changes. The second ICP start, the motion rule and the yaw-only constraint change how the frames are used. The recorded measurements stay the same.

Predictions. I wrote a prediction in each experiment before its recorded run, but I had usually tried the same settings on the server first, so they were not blind. f1_baseline was only an exploratory first pass, sealed with the label and prediction still TODO, and it has no selected run. f1_final was a real miss: I predicted 0.05-0.15 m and got 0.274 m, because RANSAC on the Mac gave a slide at pair 104 to 105. I also predicted that its selection would make things worse, but it was slightly better.

3. Results and discussion

3.1 Comparison table

The mean L2 and frame counts come from sealed experiments on my MacBook Air M1 (Open3D 0.18). Time is the total execution time for the same settings measured on one lab server (Open3D 0.20), so the rows can be compared. The server's f1_yaw selection kept 186 frames instead of 187.

Experiment / run	Floor	Voxel	Pipeline	Frames	Retained	Mean L2 (m)	Time (s)
f1_baseline / baseline	1	5 cm	RANSAC + ICP	191	100%	3.174	137
f1_final / baseline	1	8 cm	+ identity (fitness rule)	191	100%	0.274	63
f1_final / selected	1	8 cm	same, fitness >= 0.70	187	97.9%	0.268	62
f1_robust / baseline	1	8 cm	+ motion rule	191	100%	0.092	65
f1_robust / selected	1	8 cm	same, fitness >= 0.70	187	97.9%	0.083	63
f1_yaw / baseline	**1**	**8 cm**	**+ yaw-only**	**191**	**100%**	**0.074**	**68**
f1_yaw / selected	1	8 cm	same, fitness >= 0.70	187	97.9%	0.071	66
f2_robust / baseline	2	8 cm	motion rule	314	100%	0.300	55
f2_robust / selected	2	8 cm	same, yield >= 500	307	97.8%	3.442	53
f2_yaw / baseline	**2**	**8 cm**	**+ yaw-only**	**314**	**100%**	**0.163**	**58**
f2_yaw / selected	2	8 cm	same, yield >= 500	307	97.8%	3.436	56

3.2 Reconstruction screenshots

In all screenshots, the estimated path is red, and GT is black.

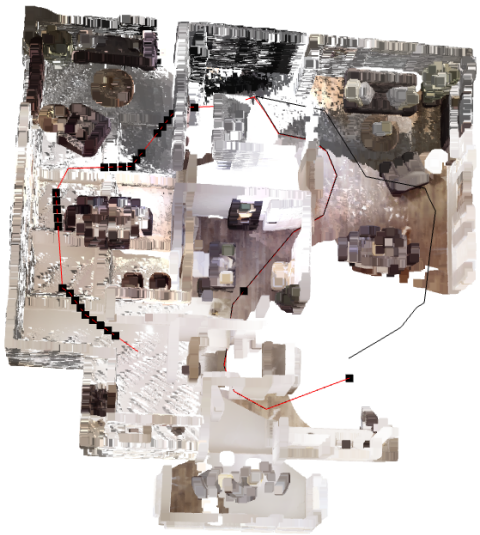


Figure 3. Floor 1, f1_baseline first pass, 5 cm voxel, 3.174 m. The paths separate after the flip.

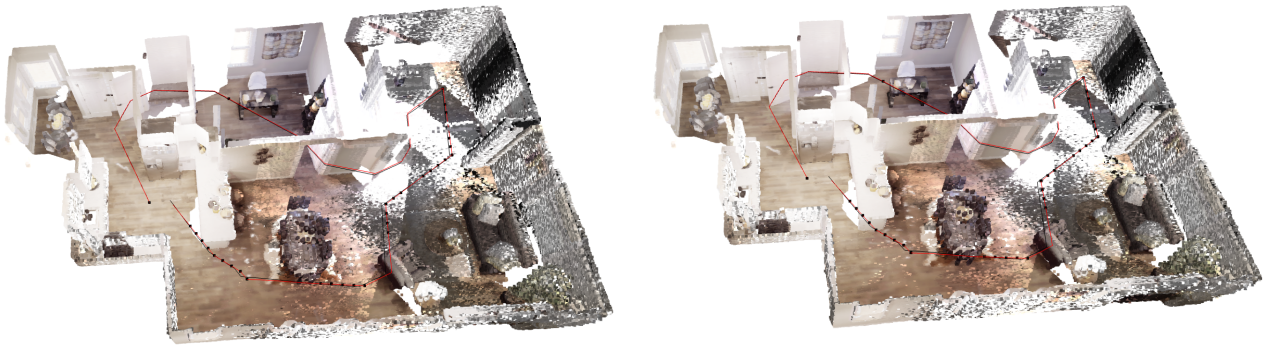


Figure 4. Floor 1 at 8 cm. Left: f1_yaw baseline, 0.074 m. Right: f1_yaw selected, 0.071 m.

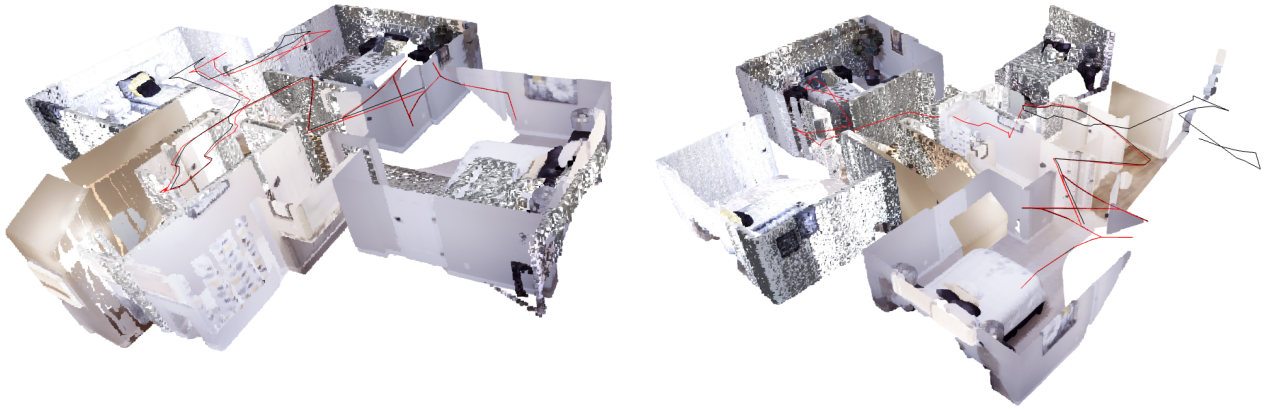


Figure 5. Floor 2 at 8 cm. Left: f2_yaw baseline, 0.163 m. Right: f2_yaw selected, 3.436 m. The paths split after the 164 to 172 gap.



Figure 6. Floor 2, f2_robust baseline, 0.300 m. The 13 degrees of roll tilted the map, so the ceiling crop missed most room ceilings.

The provided crop removes the 1 m band below the map's highest point. With yaw-only, the ceiling is level and the crop works (Figure 5).

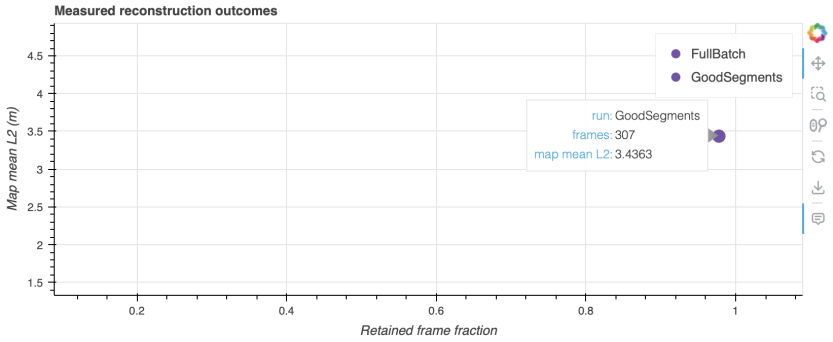


Figure 7. f2_yaw tradeoff plot. Only the selected point (3.4363 m) is inside the default axis range. The baseline point (100% retained, 0.1625 m) is below it.

3.3 Comparing the two phases

- Both floors use the same final pipeline: 0.074 m on floor 1 and 0.163 m on floor 2, which are about 0.3% and 0.5% of the 26.8 m and 32.1 m paths, respectively.
- On both floors, the big errors come from a few pairs in front of flat walls with little texture.
- Frame selection did not fix those pairs. On floor 1, it gave a small gain of 0.003-0.008 m. On floor 2 it was much worse: dropping 165-171 makes 164 to 172 one 60-degree step, which registers about 180 degrees wrong (3.44 m).
- What helped most was changing the pipeline and, on floor 2, the capture as well. Recording again with smaller turns gave a much better result than the first capture's 8.48 m.
- The remaining error on floor 2 is drift. A yaw error of 1-2 degrees at the wall grows over the remaining path of about 20 m. Loop closure would be the next step, but I did not implement it.

4. Questions

Q1. What happens if ICP is performed without global registration (RANSAC)? Why?

ICP only finds the nearest local minimum, so it needs an initial point close to the answer. Without RANSAC, the start is the identity. With my 8 cm settings on the server, ICP from identity gave 0.480 m on floor 1 (0.075 m with RANSAC) and 1.104 m on floor 2 (0.160 m), so about 6.5 times worse. Registration was about twice as fast, 16-17 s against 33-38 s. Even a small step is a big motion for far points: a 10-degree turn moves a wall 3 m away by about 0.5 m, much more than the 12 cm threshold, so ICP matches the wrong points. RANSAC with FPFH matches based on local shape, so it does not require a good initial estimate; ICP then refines the pose. RANSAC is not always right either, since it can flip on a blank wall, so my pipeline also tries the identity start. On my first floor 2 capture, both methods failed (8.48 m with RANSAC, 4.69 m with ICP only), because some pairs had almost no overlap.

Q2. Describe the choices used to improve ICP stability.

Voxel size 8 cm, because it reduces depth noise and gives stable normals, and it was the best in my tests. Correspondence threshold of 1.5 times the voxel size for RANSAC and ICP. Point-to-plane ICP, which is well-suited to scenes with many walls and floors. Mutual filter and checkers in RANSAC. A second ICP starts from an identity and uses motion size, fitness, and RMSE to reject flips and slides. The yaw-only constraint, from the agent's motion, removes the roll that a flat wall cannot measure. A fixed seed per pair, and tests over seeds 0-4 for the motion rule and 0-2 for the final pipeline (floor 1: 0.074-0.075 m, floor 2: 0.154-0.161 m on the server). I did not use a robust loss.

Q3. How do the observed data-quality factors and collection choices relate to reconstruction performance and Mean L2?

Mean L2 depends on a few weak pairs because an error in one pair propagates to all later poses. No single factor finds these pairs on both floors. High ICPInlierRMSE marked the wall on floor 1, low VoxelizedPointYield marked the wall on floor 2, and fitness marked neither. The lowest-fitness pairs on both floors were correct, and the frame with the most missing depth registered fine. Collection choices mattered a lot: large turns between saved frames broke my first capture (8.48 m), and standing close to a blank wall caused most of the remaining floor 2 error. For better capture, I would keep steps small, keep corners or furniture in view as I turn, and avoid standing close to a flat wall.